SUDARSHAN MADDI

APPLIED AI ENGINEER

CONTACT

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Woxsen University

Linkedin Profile

SKILLS

- **Programming:** Python, JavaScript, SQL, Bash
- **AI / Machine Learning:** ML, NLP, CV, Deep Learning
- **LLM & Generative AI Systems:** LLMs, RAG, Multi-Agent AI Systems, LoRA Fine-Tuning, Prompt Engineering, Embeddings, Agentic AI Systems, Vector Databases, LLM Inference Pipelines, Model Fine-Tuning, AI System Orchestration
- **Frameworks & Libraries:** LangChain, LangGraph, HuggingFace Transformers, PEFT, Streamlit, NumPy, Scikit-learn
- **Web & Backend:** Node.js, React.js, REST APIs
- **Databases:** MongoDB, Supabase
- **AI APIs:** OpenAI, LLaMA-3, OpenRouter, HuggingFace Hub
- **Version Control:** Git, GitHub

PROFILE

Applied AI Engineer and Data Science undergraduate specializing in large language model (LLM) systems, multi-agent AI architectures, and retrieval-augmented generation (RAG) pipelines. Experienced in designing and deploying production-grade AI workflows using Python, LangChain/LangGraph, and modern LLM APIs, including parameter-efficient fine-tuning of the Phi-3 model using LoRA and HuggingFace Transformers. Built end-to-end intelligent systems ranging from multi-agent knowledge retrieval platforms and LLM-driven decision support systems to real-time computer vision applications. Interested in developing scalable AI infrastructures that bridge advanced research models with real-world production environments.

WORK EXPERIENCE

LOKACHAKRA — JULY - PRESENT

Applied AI Engineer

- Architected and developed CrisisCommander.AI, an agentic decision-support platform leveraging LLaMA-3, Python, and RAG pipelines to automate crisis analysis and response generation.
- Engineered multi-agent AI architectures for the company's platform, designing specialized agents for task orchestration, knowledge retrieval, and automated reasoning workflows.
- Implemented LLM orchestration pipelines, backend APIs, and model integration workflows to enable scalable real-time decision support and production-ready AI deployment.

COGNIFYZ TECHNOLOGIES — JANUARY - FEBRUARY

Machine Learning Engineer

- Developed machine learning models for restaurant rating prediction using regression algorithms (Linear Regression, Random Forest), implementing data preprocessing, feature encoding, and model evaluation using RMSE and R^2 metrics.
- Built a content-based restaurant recommendation system using feature similarity and cosine similarity analysis, enabling Top-N recommendations based on cuisine type, location, and pricing attributes.
- Implemented a multi-class cuisine classification pipeline and performed geospatial restaurant data analysis, applying feature engineering, data visualization, and exploratory data analysis using Python ML libraries

BLUESTOCK FINTECH — DECEMBER - JANUARY

Software Developer Engineer

- Designed and implemented an automated financial analysis pipeline that retrieves company financial data from external APIs, processes key indicators, and generates structured investment insights using Python-based rule-driven analytics.
- Engineered a data processing workflow to transform raw JSON API responses into structured financial metrics such as ROE, profit growth, sales growth, and debt ratios for analytical processing.
- Integrated MySQL database storage using SQLAlchemy, enabling scalable persistence of generated company insights for backend services and web application integration.

HACKATHONS

- Airothon 2025 – Agentic AI Hackathon: Secured 4th place and ₹1,500 prize; developed multi-agent AI prototypes for autonomous decision-making.
- Swafinix AI Agents Hackathon: Ranked 10th out of 2,664 participants; designed and implemented agent-based AI solutions.
- Swasthathon 2025: Top 14 finalist team; built AI-driven health monitoring and automation solution.
- NextGen Hackathon 2025: Finalist; contributed to innovative AI system prototypes.

ACTIVITIES

- CODE{X} - THE PROGRAMMING CLUB: Core Team (PRESENT)
- INTERNATIONAL RELATIONS COMMITTEE (PRESENT)
- IGNITE - ENTERPRENUERSHIP CLUB - EXECUTIVE (2025)

EDUCATION

Woxsen University

School of Technology

Hyderabad , India

2024 - 2028

Excellencia Junior College

Hyderabad, India

Overall Grade : 93%

2022-2024

INFRYNTREK SYSTEMES

JANUARY - FEBRUARY

Data Analyst Intern

- **Performed fraud detection analysis on financial transaction datasets, applying exploratory data analysis, feature engineering, and classification modeling (Logistic Regression) to identify patterns with fraudulent activity.**
- **Implemented customer segmentation using RFM analysis and K-Means clustering, enabling behavioral grouping of users and estimation of Customer Lifetime Value (CLV) for targeted business strategies.**
- **Developed loan default risk prediction models using machine learning classification techniques and evaluated model performance using metrics such as precision, recall, and ROC-AUC.**

PROJECTS

DebugLLM — Production LLM Code Debugging System

Phi-3 Mini · LoRA (PEFT) · vLLM · FastAPI · Redis · Docker · Modal

- Engineered a production LLM debugging system using LoRA-fine-tuned Phi-3 Mini with vLLM inference, Redis caching, and INT4 quantization to enable automated Python bug detection and low-latency code correction.
- Built scalable inference infrastructure using vLLM, Redis caching, and API orchestration, enabling low-latency model serving and efficient GPU inference.
- Integrated Langfuse observability and monitoring pipelines to track prompt performance, inference latency, and debugging accuracy across model responses.

GitHub Multi-Agent RAG — AI Codebase Intelligence System

Python · LangGraph · FAISS · OpenRouter · Streamlit

- Designed a multi-agent RAG architecture using LangGraph and FAISS to perform codebase-level repository analysis and generate automated documentation insights.
- Implemented repository ingestion, file chunking, and embedding generation pipelines to enable semantic search across large codebases.

Academia AI — Multi-Agent Research Synthesis System

Python · LangGraph · FAISS · Streamlit · OpenRouter

- Built an AI research assistant that processes academic papers using multi-agent orchestration and RAG pipelines to generate structured research summaries.
- Implemented semantic document chunking, embedding generation, and retrieval workflows for automated research insight synthesis.

Gesture-OS — Real-Time Gesture-Controlled Computer Interface

Python · OpenCV · MediaPipe · PyAutoGUI · NumPy

- Implemented a real-time hand-tracking pipeline using OpenCV and MediaPipe to enable gesture-based cursor control and OS command automation.
- Integrated gesture classification and OS command automation using MediaPipe landmark detection and PyAutoGUI control pipelines.

AiroClaim — AI Insurance Claim Processing System

Python · Streamlit · EasyOCR · PyMuPDF · Pandas

- Built an AI workflow that extracts structured claim data from documents using OCR-based text extraction and parsing pipelines.
- Implemented rule-based fraud detection and structured data extraction pipelines to automate insurance claim evaluation.

AstroRisk — Space Weather Risk Analytics Platform

Python · Streamlit · Space Weather APIs

- Developed a platform analyzing solar storm and space weather data to estimate infrastructure risks affecting satellites, aviation, and power grids.
- Implemented real-time data ingestion, risk scoring models, and interactive dashboards for domain-specific analytics.